[Thm. Let $R_0 > 0$ be given.
Suppose that f is analytic on the upper-half plane outside of $R_0 e^{i\theta}$ $(0 \le \theta \le \pi)$.
and for $C_R : t \mapsto R \cdot e^{it\theta}$ $(0 \le \theta \le \pi)$ where $R > R_0$, f satisfies

there is M_R such that $\sup_{z \in C_R} |f(z)| \le M_R$ for all $R > R_0$

and $\lim_{R \to \infty} M_R = 0$

Then, for all $a > 0$,

$$\lim_{R \to \infty} \int_{C_R} f(z) \cdot e^{iaz}\, dz = 0.$$

Proof. First, we will show the Jordan inequality:
Since $\sin x \ge \frac{2}{\pi} \cdot x$ for all $x \in [0, \frac{\pi}{2}]$, therefore

$$e^{-R \cdot \sin\theta} \le e^{-R \cdot \frac{2}{\pi} x} \quad \text{for all } x \in [0, \tfrac{\pi}{2}].$$

Then e^{-cx} is integrable explicitly, so

$$\int_0^{\frac{\pi}{2}} e^{-R \cdot \sin\theta}\, d\theta \le \int_0^{\frac{\pi}{2}} e^{-R \cdot \frac{2}{\pi} x}\, dx = \frac{\pi}{2R} \cdot \left(1 - e^{-R}\right) \le \frac{\pi}{2R}.$$

So, by the symmetricity of $e^{-R \cdot \sin\theta}$, we obtain the Jordan inequality

$$\int_0^{\pi} e^{-R \cdot \sin\theta}\, d\theta \le \frac{\pi}{R}. \quad \text{Jordan's inequality.}$$

Now,

$$\int_{C_R} f(z) \cdot e^{iaz}\, dz = \int_0^{\pi} \underbrace{f(Re^{ix})}_{|\cdot| \le M_R} \cdot \underbrace{e^{ia \cdot Re^{ix}}}_{|\cdot| \le e^{-aR\sin x}} \cdot R \cdot ie^{ix}\, dx$$

, so that

$$\left| \int_{C_R} f(z) e^{iaz}\, dz \right| \le M_R \cdot R \cdot \int_0^{\pi} e^{-aR\sin x}\, dx < \frac{M_R \pi}{a} \to 0 \text{ as } R \to \infty.$$

□